



Sanjivani Rural Education Society's

Sanjivani College of Engineering, Kopargaon-423603

(An Autonomous Institute Affiliated to Savitribai Phule Pune University, Pune)

NAAC 'A' Grade Accredited, ISO 2001:2018 Certified

Department of Information Technology

NBA Accredited-UG Programme

Class: B. Tech. Semester: VII
Subject: Advanced Data Science Laboratory (PEIT412B)

Practical No.: 01

CO1: Implement deep learning models using TensorFlow/PyTorch frameworks. (BTL-Apply)

Title:

Assignment based on implementation of Artificial Neural Network (ANN) using TensorFlow/Keras for prediction problems.

Software Requirements:

- Python (Recent Version), TensorFlow/Keras, Pandas, Scikit-learn.
- Visual Studio Code (or any other editor), Jupyter Notebook / Google Colab.

Learning Outcomes:

By the end of this assignment, students should be able to:

- Understand the basic architecture and working of an Artificial Neural Network (ANN).
- Build, compile, and train ANN models using TensorFlow/Keras.
- Apply ANN models to solve both regression and classification prediction problems.
- Understand the role of activation functions, loss functions, and optimizers.
- Evaluate ANN model performance using appropriate metrics (MAE, RMSE, Accuracy, F1-Score).

Exemplars / Case Study: Implement ANN models for the following prediction problems:

- **Student Performance Prediction:** Predict a student's marks/grade (regression) using features such as study hours, attendance, and past scores.
- **House Price Prediction:** Predict house prices (regression) based on features like area, location, number of rooms, and age of property.
- **Employee Attrition Prediction:** Predict whether an employee will leave the company (binary classification) using features like salary, satisfaction level, and years at company.
- Use Pandas for data ingestion, cleaning, and pre-processing (handling missing values, encoding, scaling).
- Build an ANN using the Keras Sequential API with appropriate input, hidden, and output layers.
- Compile the model with a suitable loss function and optimizer, then train it using `model.fit()`.
- Evaluate the trained model using regression metrics (MAE/RMSE) or classification metrics (Accuracy, Confusion Matrix, F1-Score) as applicable.

Theory:

- **Artificial Neural Network (ANN):** A computational model inspired by the human brain, made up of interconnected "neurons" arranged in layers, capable of learning complex non-linear patterns from data.

- **Structure of ANN:** Consists of an Input Layer (receives features), one or more Hidden Layers (learn patterns), and an Output Layer (produces the prediction).
- **Neuron/Perceptron:** Basic unit that computes a weighted sum of inputs, adds a bias, and passes the result through an activation function.
- **Activation Functions:** Introduce non-linearity so the network can learn complex relationships. Common types: ReLU (hidden layers), Sigmoid (binary classification output), Softmax (multi-class output), Linear (regression output).
- **Forward Propagation:** Process of passing input data through the network layer by layer to generate a prediction.
- **Backpropagation:** Algorithm that calculates the error at the output and propagates it backward to update weights using gradient descent, so the model learns.
- **Loss Function:** Measures the difference between predicted and actual values. E.g., Mean Squared Error (MSE) for regression, Binary/Categorical Cross-entropy for classification.
- **Optimizer:** Algorithm (e.g., SGD, Adam, RMSprop) that adjusts model weights to minimize the loss function.
- **Epochs & Batch Size:** An epoch is one full pass over the training data; batch size is the number of samples processed before the model's weights are updated.
- **Overfitting & Dropout:** Overfitting occurs when a model memorizes training data instead of generalizing; Dropout randomly disables neurons during training to reduce overfitting.

Basic Code Info (Sample Skeleton using TensorFlow/Keras):

A generalized ANN pipeline that can be adapted for Student Performance, House Price, or Employee Attrition prediction:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# 1. Data Ingestion
df = pd.read_csv('dataset.csv')
df = df.dropna()          # handle missing values

# 2. Feature/Target split
X = df.drop('target', axis=1)
y = df['target']

# 3. Train-test split & scaling
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 4. Build the ANN
model = Sequential([
    Dense(32, activation='relu', input_shape=(X_train.shape[1],)),
    Dense(16, activation='relu'),
```

```

Dense(1, activation='linear') # 'sigmoid' for binary classification
])

# 5. Compile the model
model.compile(optimizer='adam',
              loss='mse',          # 'binary_crossentropy' for classification
              metrics=['mae'])     # 'accuracy' for classification

# 6. Train the model
history = model.fit(X_train, y_train, epochs=50,
                   batch_size=16, validation_split=0.2)

# 7. Evaluate the model
results = model.evaluate(X_test, y_test)
print('Test Loss, Test MAE:', results)

```

Note: For Employee Attrition (classification), change the output layer to Dense(1, activation='sigmoid'), the loss to 'binary_crossentropy', and evaluate using accuracy, confusion_matrix, and classification_report from sklearn.metrics.

Frequently Asked Questions:

- What is an Artificial Neural Network, and how does it mimic biological neurons?
- What is the role of activation functions in an ANN? Explain ReLU, Sigmoid, and Softmax.
- Explain the difference between forward propagation and backpropagation.
- How does the choice of output layer activation differ for regression vs. classification problems?
- What are loss functions and optimizers, and how do you choose the right one?
- How do epochs, batch size, and learning rate affect model training?
- What is overfitting in ANN, and how does Dropout help prevent it?
- How do you evaluate ANN model performance differently for regression and classification tasks?

Conclusion:

Thus, we have implemented the assignment based on Artificial Neural Network (ANN) using TensorFlow/Keras for prediction problems, covering Student Performance Prediction, House Price Prediction, and Employee Attrition Prediction.

Text Books:

1. Joseph Babcock & Raghav Bali, Generative AI with Python and TensorFlow: Build Intelligent Generative AI Models, Packt Publishing, 2024, ISBN-10: 1807780211, ISBN-13: 9781807780210.
2. Chip Huyen, AI Engineering: Building Applications with Foundation Models, O'Reilly Media, 2025, ISBN-10: 1098166302, ISBN-13: 9781098166304.
3. Mariprasath Dhanasekaran, Cheepati & Rivera, Practical Guide to Machine Learning, NLP, and Generative AI: Libraries, Algorithms and Applications, River Publishers, 2025, ISBN-13: 9788770046534.

4. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, 3rd Edition, O'Reilly Media, 2022, ISBN-10: 9355421987, ISBN-13: 9789355421982.

Reference Books:

1. Lewis Tunstall, Leandro von Werra & Thomas Wolf, Natural Language Processing with Transformers: Building Language Applications with Hugging Face, O'Reilly Media, 2022, ISBN-10: 1098103246, ISBN-13: 9781098103248.
2. Aston Zhang, Zachary C. Lipton, Mu Li & Alexander J. Smola, Dive into Deep Learning, Cambridge University Press, 2021, ISBN-10: 1009389432, ISBN-13: 9781009389433.

**Prepared By,
Mr. U. B. Sangule
Subject In-charge**

**Approved By,
Dr. M. A. Jawale
HOD-IT**